

A hybrid string-loop model: Emergent quantum gravity through iterative reverse simulation

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Zusammenfassung

String theory (ST) and loop quantum gravity (LQG) offer complementary approaches to quantum gravity, but have limitations in initial conditions and testability. This paper proposes the String-Loop Emergent Framework (SLEF): a hybrid model that integrates STs vibrating strings as “dynamic loops” in LQGs spin networks. Through an iterative reverse simulation – backwards from the observed universe – we reduce the state space exponentially to derive emergent parameters (e.g. 5 primordial constants). Numerical prototypes (Python-based) demonstrate a reduction in complexity by 10^{50} factors. Predictions include testable CMB anisotropies and modified big bounce dynamics. SLEF solves fine-tuning emergently and connects ST’s landscape with LQG’s discreteness.

Keywords: String theory, loop quantum gravity, hybrid model, reverse simulation, emergent quantum gravity, fine-tuning

1 Introduction

The apparent fine-tuning of the fundamental constants of the universe, as highlighted in discussions of the anthropic principle [1], remains a mystery. String theory (ST) promises a theory of everything, but is hampered by its immense vacuum landscape. Loop quantum gravity (LQG) quantizes space-time discretely, but struggles with the diversity of initial conditions. This paper presents the String-Loop Emergent Framework (SLEF), a hybrid model that combines both approaches and reduces the state space through iterative reverse simulation. Inspired by 5 primordial parameters (E, g, S, Y, Φ), SLEF fine-tuning solves emergently.

2 Theoretical framework

2.1 String theory elements

The ST models particles as strings in 10 dimensions. The Polyakov action is:

$$S = -\frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{-h} h^{ab} \partial_a X^\mu \partial_b X^\nu G_{\mu\nu}(X), \quad (1)$$

expanded to include primordial coupling g as scaling.

2.2 LQG-Elemente

LQG quantizes via spin networks. The constraint:

$$\hat{H}\Psi[\gamma, \vec{A}] = 0, \quad (2)$$

with wave functional Ψ via connections $\vec{A}$.

2.3 Complementarity

SLEF integrates strings as vibrating edges. The hybrid Hamiltonian:

$$H_{SLEF} = H_{LQG}(\Psi_n) + \sum_{j=1}^n L_{string}(E, g, S, Y, \Phi) \cdot V_{loop,j}. \quad (3)$$

Derivation: 1. LQG-Basis: $H_{LQG} = \int d^3x N \left(\frac{E_i^a E_j^b}{\sqrt{\det q}} \epsilon^{ijk} F_{abk} + \dots \right)$. 2. ST integration: $L_{string} = -\frac{1}{4\pi\alpha'} g \partial_\sigma X^\mu \partial_\sigma X_\mu$. 3. Sum over loop volumes $V_{loop,j} \approx \sqrt{j(j+1)} \ell_P^3$. Sum over loop volumes:

$$\frac{\partial H_{SLEF}}{\partial \Psi_n} = \frac{\partial H_{LQG}}{\partial \Psi_n}. \quad (4)$$

3 Das SLEF-Modell

3.1 Modellbeschreibung

SLEF: Discrete loops with string oscillations. Effective metric:

$$g_{\mu\nu}^{eff} = g_{\mu\nu}^{LQG} + \delta g_{\mu\nu}^{string} = \sum_j V_{loop,j} \cdot (\partial^\mu X^\nu + \Phi \cdot \epsilon^{\mu\nu\rho\sigma} \partial_\rho X_\sigma). \quad (5)$$

3.2 Iterative reverse simulation

Inverse transformation:

$$\Psi_{n-1} = f^{-1}(\Psi_n, \mathbf{\Pi}). \quad (6)$$

Derivation: 1. Vorwärts: $\Psi_n = e^{-iH_{SLEF}\Delta t}\Psi_{n-1}$. 2. Inverse: $\Psi_{n-1} = e^{iH_{SLEF}\Delta t}\Psi_n$, with phases:

$$f^{-1}(\Psi_n, E, g, S, Y, \Phi) = \Psi_n \cdot \exp\left(i \int g \cdot S \cdot dV_{loop} + Y \cdot \Phi \cdot \partial_t \Psi_n\right). \quad (7)$$

3. Filtering: Likelihood $P(\Psi_k|\text{Daten}) \propto \exp(-\frac{1}{2}\chi^2(\Psi_k))$. 4. Convergence: $\dim(\mathcal{H}_N) \approx \exp(-N \cdot \lambda)$, $\lambda \approx 0.5$. Partielle:

$$\frac{\partial \Psi_{n-1}}{\partial \Psi_n} = I + i\Delta t \frac{\partial H_{SLEF}}{\partial \Psi_n}. \quad (8)$$

3.3 Integration of the primordial parameters

Spin-Labels: $j_l = \lfloor g \cdot S \cdot E + Y \cdot \Phi \cdot n \rfloor$.

4 Numerical simulations and results

4.1 Prototype implementation

Extension of Grok Physics Explorer: Finite differences for H_{SLEF} .

4.2 Results

[Placeholders for mappings: state space reduction, homogeneity metric.]

4.3 Validierung

Comparison with CMB data.

5 Predictions and implications

Power spectrum: $P(k) = P_{LQC}(k) \cdot (1 + g \cdot \delta_{string}(k))$. Testable: CMB anisotropies, 1 TeV scalar.

6 Discussion and limitations

Advantages: Testability. Limitations: computational effort.

7 Conclusions

SLEF as a promising hybrid.

Literatur

- [1] Josef M. Gaßner. Warum ist die welt so wie sie ist? anthropisches prinzip.
<https://www.youtube.com/watch?v=example>, February 2024.